Vikramaditya Rajput

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Education

University of Illinois at Urbana-Champaign Champaign, IL

Bachelor of Science in Computer Science and Statistics | GPA: 3.95/4.0 *Expected Graduation: May 2028*

- **Relevant Coursework**: Data Structures & Algorithms, Computer Architecture, Numerical Methods, Discrete Structures, Statistics and Probability Theory, Applied Linear Algebra

Experience

Software Development Intern May 2026 – Aug 2026

Tradegrub *Lewes, DE*

∗ Engineered dual AI-native MCP servers using FastAPI to expose GraphQL/REST APIs to LLMs, reducing campaign setup time by 80%; secured endpoints with stateless JWT auth and a four-tier RBAC hierarchy.
∗ Architected a dynamic slot-generation algorithm and database-agnostic SQLAlchemy 2.0 data layer, preventing double-bookings via atomic transactions while processing 10k+ daily availability checks at <45ms p99 latency.
∗ Built an event-driven analytics pipeline merging 7 social APIs, Stripe, and Firebase via HMAC-verified webhooks, implementing token-bucket rate limiters to ensure reliable data ingestion for ROI and retention tracking.

Eco Illini Supermileage Sep. 2025 – Present

Software and Data Lead

∗ Developed an Extended Kalman Filter (EKF) in MATLAB for vehicle state estimation, utilizing Particle Swarm Optimization (PSO) to tune model parameters and targeting a 15–20% reduction in positional noise.
∗ Simulated GPS data and modeled motor dynamics to calculate real-time power consumption, analyzing over 10,000 data points per run to identify potential energy efficiencies.
∗ Analyzed telemetry outputs across 50+ simulated race profiles to make optimized driver strategy plans.

Eagle Scout Mar. 2018 – Dec. 2024

Boy Scouts of America

∗ Improved community spaces for the Animal Alliance of New Jersey by constructing 6 outdoor structures, actively recruiting and directing a volunteer team of 15 scouts through a multi-phase design and build process.
∗ Oversaw full project execution, managing design selection, materials procurement, work-session scheduling, construction oversight, and final transport/installation of all structures.

Projects

Limit Order Book Reconstruction Engine | *C++20, CMake, Python, Nasdaq ITCH 5.0* May 2026 – Present

∗ Built a single-threaded L3 order book engine that reconstructs full market depth from raw Nasdaq ITCH 5.0 binary feeds, handling add/execute/cancel/delete/replace semantics across 30M+ messages per trading day per ticker with zero-copy parsing and a custom intrusive price-level structure.
∗ Achieved 1.5M messages/sec throughput at 55µs p99 per-update latency, a 2× speedup over a std::map baseline, by replacing node-based containers with a flat array-indexed price ladder and an open-addressing order ID hash map; validated against exchange-published end-of-day summaries.
∗ Implemented a queue-position-aware fill simulator modeling latency, partial fills, and cancel-ahead behavior, showing that naive mid-price backtests overstated strategy PnL by 80% on 150 days of NVDA/SPY data.

Scalable ML Classification of Tree Species using Satellite Imagery May. 2025 – Aug. 2025

∗ Engineered a scalable ensemble Machine Learning (Random Forest) algorithm for tree species classification using multispectral satellite imagery, enabling analysis across large, geologically diverse regions.
∗ Enhanced model accuracy by integrating Object-Specific Attributes (OSA) such as crown ratio, tree height, and diameter; demonstrated a mean classification accuracy increase from 66.4% to 90.2% after augmentation.

Technical Skills

Languages: Python, Java, C/C++, Rust, SQL, JavaScript, Typescript, HTML/CSS, Dart, Verilog, R
Frameworks: React, Node.js, Flask, JUnit, Material-UI, Tailwind, FastAPI, Astro, Flutter
Developer Tools: Git, Docker, Google Cloud Platform, CI/CD (Github Actions)
Data/Web3: Stripe API, Pandas, Numpy, MatPlotLib, PyTorch